Solidigm™ D5-P5336 Gen4 NVMe

MLPerf Storage v2.0 Submission

System Name: Solidigm D5-P5336 Gen4 NVMe

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Submission Date: July 4, 2025

Benchmark Version: v2.0

Solution Type: Local Storage

Submission Division: CLOSED

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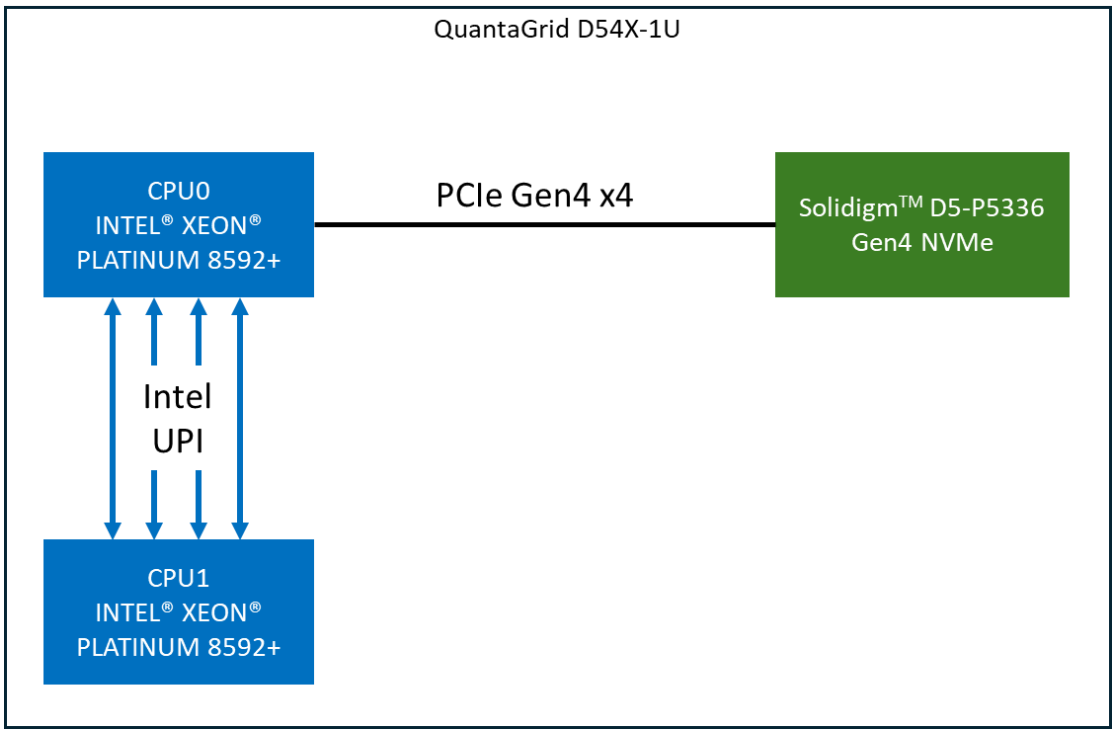
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1. Hardware

The hardware setup uses a 1U single server with locally installed NVMe drives. In the system topology, the NVMe is directly connected to CPU0 via a PCIe Gen4 x4 link.





Power Requirement Table (Direct-attached media)

System components	Power supply unit	Nameplate rated power	Design Power
Solidigm™ D5-P5336 Gen4 NVMe	Active	21W	21W
Totals		**21W**	**21W**

2. Software

The distribution is Rocky Linux release 9.5 (Blue Onyx), with kernel version 5.14.0-503.14.1.el9_5.x86_64. The UCX library required by OpenMPI is provided by the built-in ucx-1.18.0 included in NVIDIA DOCA-OFED-2.9.2. The disk file system is formatted as XFS, using the default parameters for mkfs.xfs.

3. Settings

The system itself was not heavily customized, and only the following basic configurations were applied:

- **BIOS Settings**
  - Set the CPU profile to **High Performance**
  - **Disabled CPU SNC** (resulting in a single NUMA domain per CPU)
- **Operating System Settings**
  - Enabled tuned and set the profile to **throughput-performance**

```
sudo tuned-adm profile throughput-performance
```

  - Disabled kernel parameter "**numa_balancing**"

```
sudo sysctl -w kernel.numa_balancing=0
```